

INTRODUCTION TO TABLEAU

Introduction to Data Visualization - Lecture 19

What is Tableau

Tableau is a data visualization tool that lets us analyze virtually any type of structured data and produce highly interactive and attractive graphs, dashboards, and reports in minutes.

Using Tableau Effectively

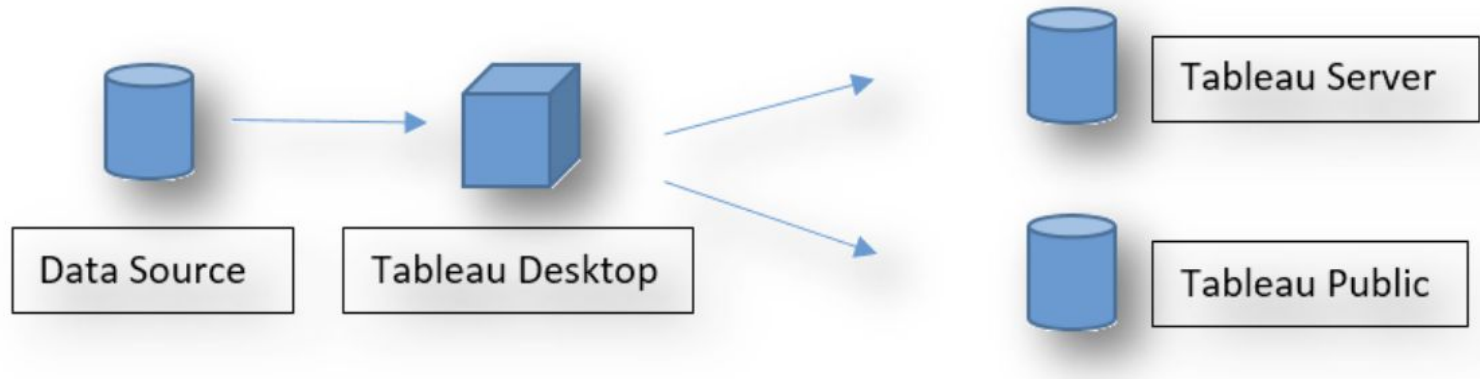
Tableau makes it easier to create powerful, visual information that communicates what is important better than a spreadsheet or text table. Tableau has advanced capabilities for more technical users, but it dramatically lowers the bar for creating dashboards and performing analytical analysis for non-technical analysts and information consumers.

INTRODUCTION TO TABLEAU

Tableau Desktop is central to the creation of all Tableau visualization and data sources. In fact, if you only want to create reports but do not have a need for circulation or wider consumption, the desktop version would be all you need.

Tableau Server is used when you have a visualization that has been designed to be consumed by others. If you have a specific, restricted audience and would like to control the manner in which they interact with your work, they should be defined on the server and you would publish your work to be accessed.

INTRODUCTION TO TABLEAU



INTRODUCTION TO TABLEAU

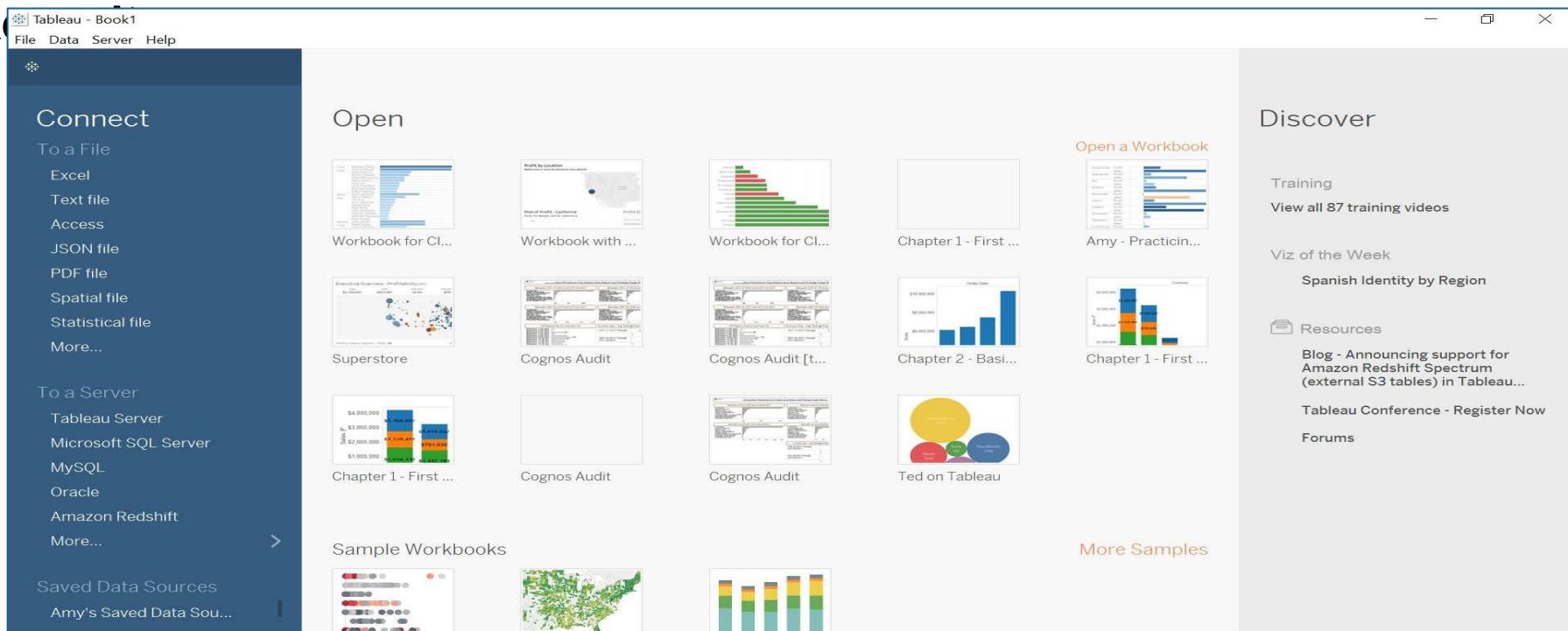
Tableau Public is a version of server but security has been setup such that the visualizations published here may be accessed without authentication. This is where you would publish work that you want anyone at all to view and consume. An example of this would be a graph demonstrating historical applications to the university.

When working in a company, they issue the Tableau Desktop License centrally and maintain their own Servers for Development, Production, and Public.

GETTING YOUR DATA READY

Choose a connection or an existing workbook from the Tableau Welcome Page.

Note: Upon installing Tableau, check your local drive for **My Tableau**



GETTING YOUR DATA READY

Once you choose your data source, you (and it) are brought to the Data Source Page where you can format your metadata.



Sheets on the left navigation pane behave like tables in a database.

GETTING YOUR DATA READY

Connections [Add](#)

Vital Stock Supplies
Excel

Sheets

- Orders
- People
- Returns
- New Union

Orders+ (Vital Stock Supplies)

Connection: ☒ Live ☐ Extract

Filters: 0 [Add](#)

Orders

People

Returns

Join

Inner Left Right Full Outer

Data Source Returns

Sort fields: Order ID = Order ID (Ret...

☐ Show aliases ☐ Show hidden fields 1,000 rows

#	Orders	Orders	Orders	Orders	Orders	Orders	Orders	Orders	Orders
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer ...	Customer ...	City	State
1	CA-2016-152...	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Claire Gute	Henderson	Kentu
2	CA-2016-152...	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Claire Gute	Henderson	Kentu
3	CA-2016-138...	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Hu	Darrin Van Hu	Los Angeles	Califo
4	US-2015-108...	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donne	Sean O'Donne	Fort Lauderda...	Florid
5	US-2015-108...	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donne	Sean O'Donne	Fort Lauderda...	Florid
6	CA-2014-115...	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoff...	Brosina Hoff...	Los Angeles	Califo
7	CA-2014-115...	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoff...	Brosina Hoff...	Los Angeles	Califo

[Go to Worksheet](#)

Data Source Sheet 1

Context Menu:

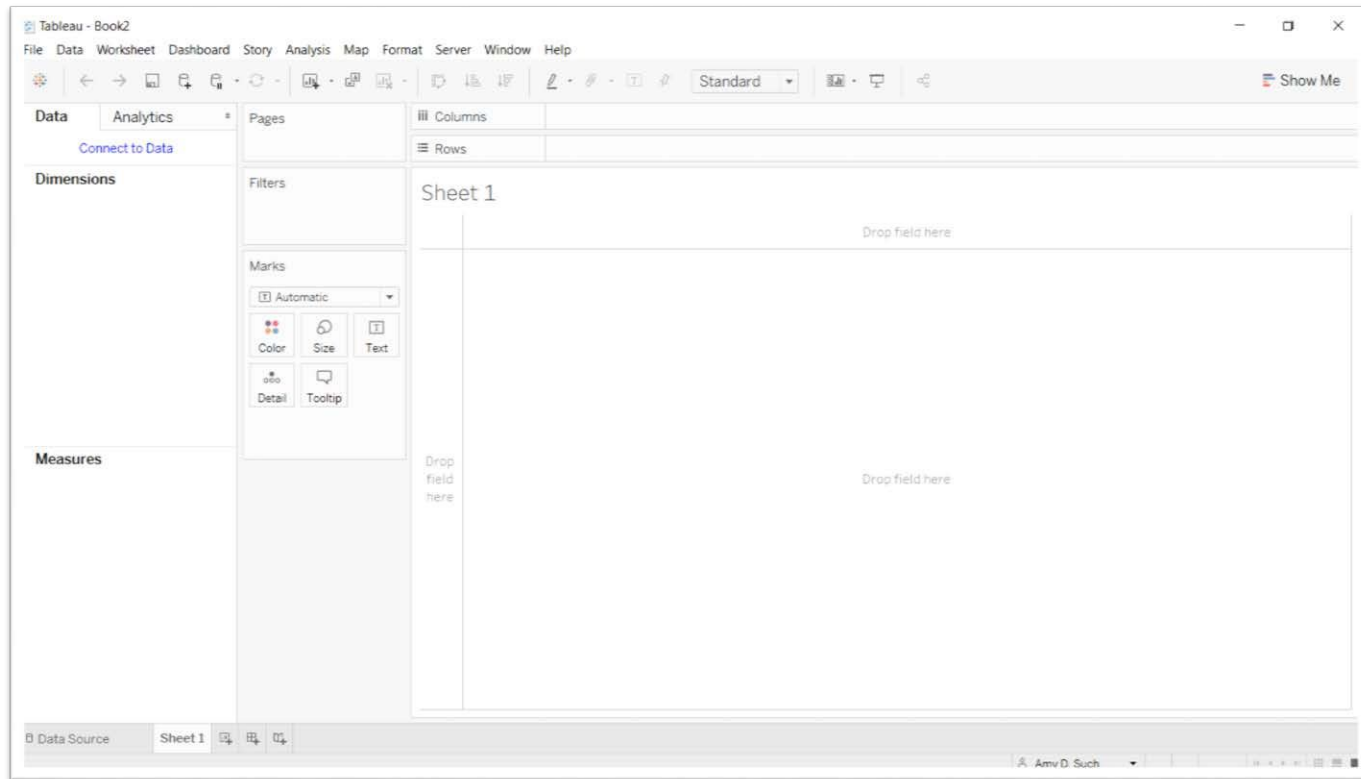
- Rename
- Copy Values
- Hide
- Aliases...
- Create Calculated Field...
- Create Group...
- Split
- Custom Split...
- Pivot (select multiple fields)
- Describe...

GETTING YOUR DATA READY

- Splitting
 - String fields can be split into multiple fields for easier analysis
 - Automatic or custom split options
- Aliasing
 - Roles (time, ship date/order date)
 - Binning (high/low sales)
- Renaming
- Data types
- Geographic roles
- Calculated fields

Data type icons in Tableau

Icon	Data type
Abc	Text (string) values
	Date values
	Date & Time values
#	Numerical values
T F	Boolean values (relational only)
	Geographic values (used with maps)



THE

INTERFACE

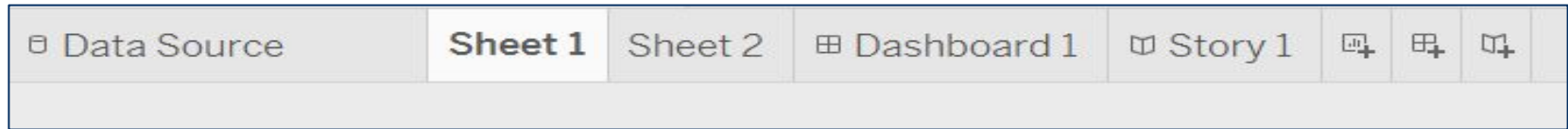
- We work in workbooks which contain worksheets, dashboards, and stories
- Worksheets are also known as **Views** of your data
- We drag and drop fields from the data source onto **Shelves**
 - Rows, Columns, Filters ...
- Items that appear on shelves are called **Pills**
 - If the Pill is **blue**, it is a Dimension
 - If the Pill is **green**, it is a Measure

iii Columns	SUM(Sales)	
≡ Rows	Region	Category

When you bring your data into Tableau, it automatically places the fields into one of two categories. Dimensions or Measures.

- **Dimensions** are fields that organize your data into categories (or buckets).
 - Individual dimension values are called **Members**
 - Usually Non-numerical and provide critical, contextual meaning to a **Measure**.
 - Who, what, when, where, why
- **Measures** return numeric values for “measuring” different dimensions.
 - Usually aggregated (sum, average, min, max, etc.)
 - Things you can do math on that are generally meaningless without context

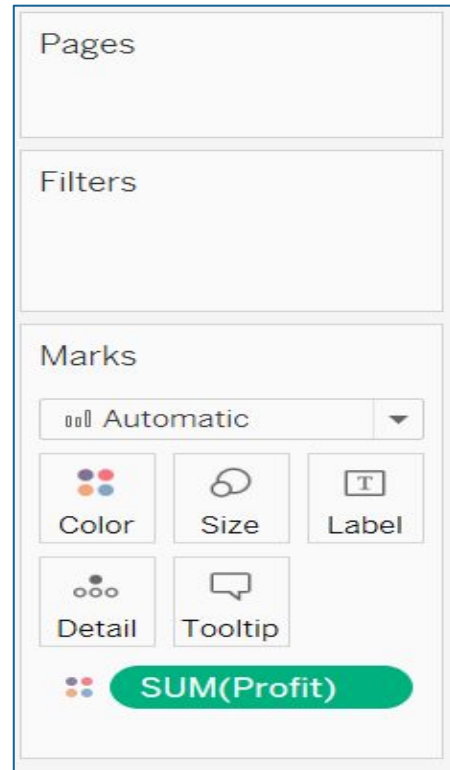
- Change between data source, worksheets, and dashboards using the tabs at the bottom
- You can also add new worksheets, dashboards, or story boards using the “new” buttons.



Once you've connected to data and evaluated available dimensions and measures in the Data pane, you'll need to decide where to drag desired dimensions and measures. You may choose to drag directly onto the visualization or onto a particular shelf or card.

Use cards to navigate pages, filters, and marks on a worksheet.

- The **Page** shelf acts as a modified Filters shelf allowing you to page through values.
- The **Filters** shelf allows you to narrow data down.
- The **Marks** card allows you to customize the display of the 'marks' that make up your chart.



Use dashboards to organize and interact with multiple worksheets

Dashboard

Layout

Device Preview

Size

Automatic

Sheets

Sales/Profit by State

Sub-Category Profit

Segment Comparison

Objects

Horizontal

Image

Vertical

Web Page

A

Text

Blank

Tiled

Floating

☒ Show dashboard title

Tiled Dashboard

Sales/Profit by State

© OpenStreetMap contributors

United States

Year of Order Date

☒ (All)
☒ 2011
☒ 2012
☒ 2013
☒ 2014

Sub-Category Profit

Machines Supplies
ChairsStorage PhonesPaper
BindersAccessoriesArt Copiers
Furnishings Fasteners
EnvelopesLabels
Bookcases

Segment Comparison

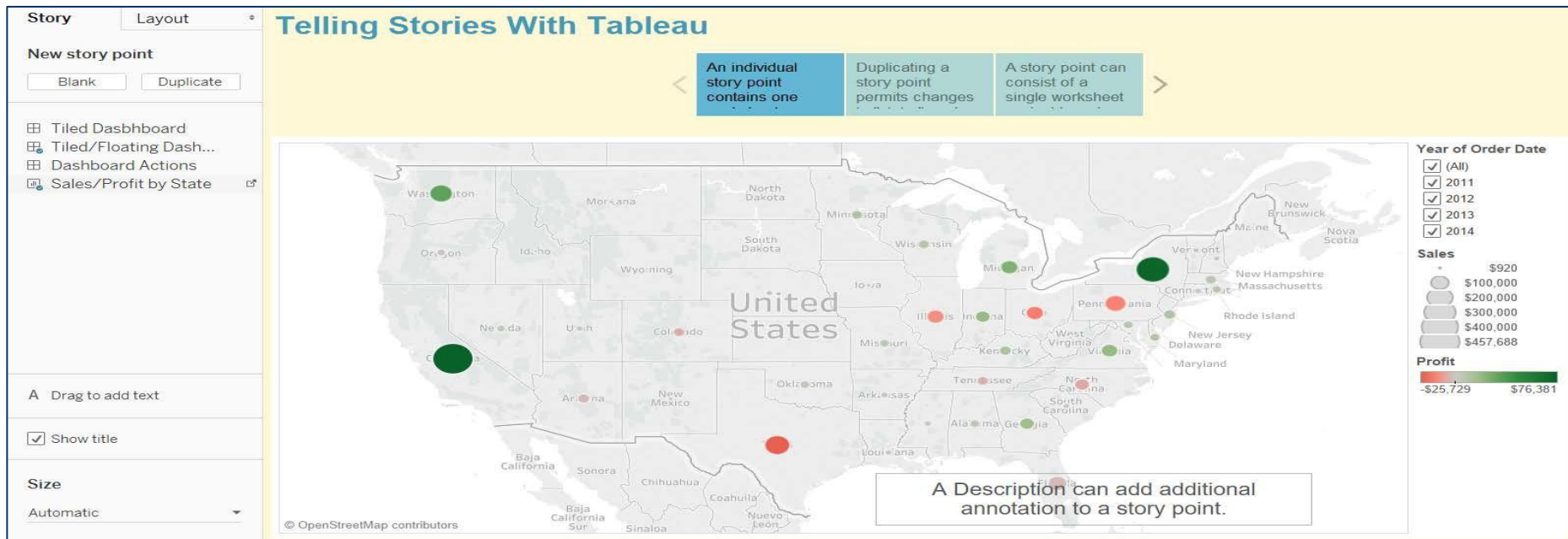
18.70%
Home Office

50.56%
Consumer

30.74%
Corporate

Note: Worksheets within dashboards are hidden from the tab list at the bottom of the screen. To re-display them, right-click on a worksheet in the dashboard window to the left and uncheck Hide Sheet.

Use stories to combine elements and create directed, interactive analysis



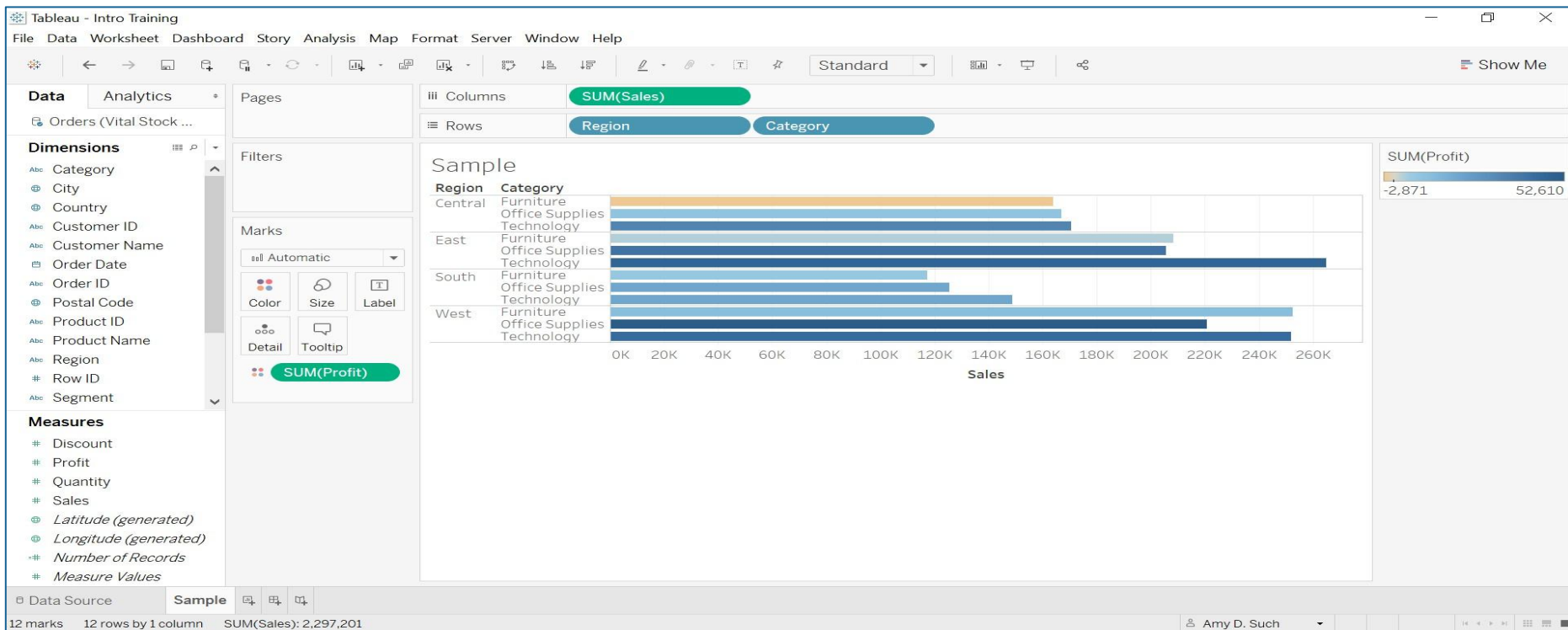
Anytime you're editing a new or existing worksheet, you'll notice the Show Me tab at the upper right of the screen.

- Show Me displays a series of thumbnail images representing the different types of charts you can create with just a few clicks.
- If you've already created a chart but would like to switch to another, clicking a new thumbnail from the Show Me dialog box will change your current visualization.

Note: Only thumbnails for visualizations that are applicable to the dimensions and measures you've selected from your data pane will be active.



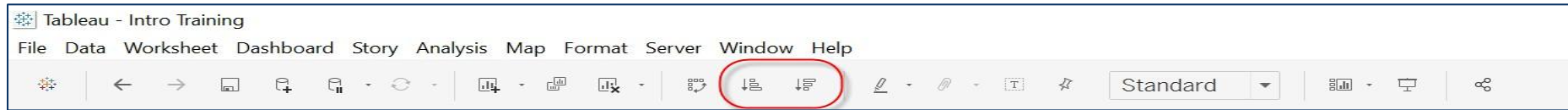
Add Dimensions and/or Measures from the Data pane to the Shelves and Cards to create visualizations.



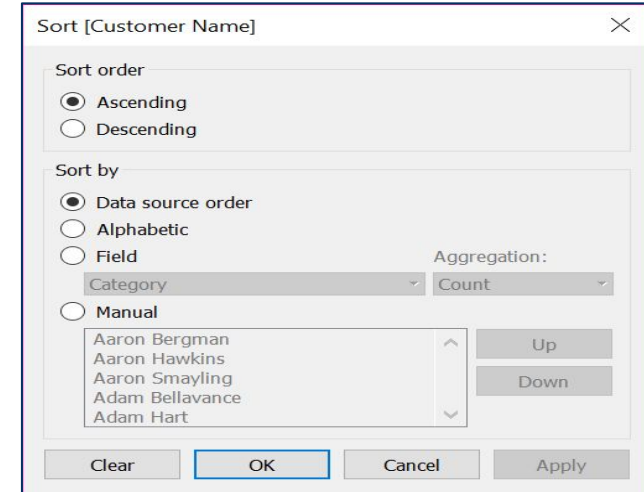


**SORTING &
FILTERING**

- Analysis can be enhanced by sorting results
- Sorting can be done using buttons for simple sorting

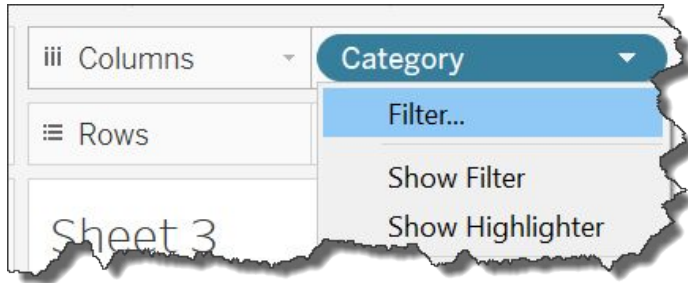


- Or ... manually through the advanced sorting dialog which is achieved by right-clicking on a Pill from the Shelf and clicking **Sort**.

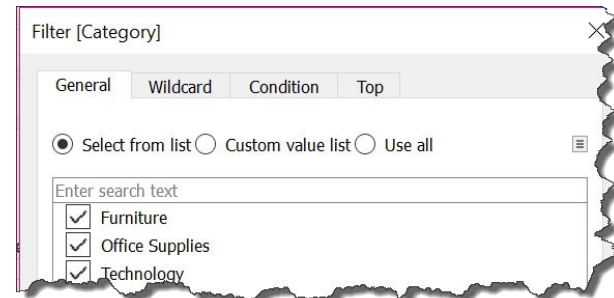


SORTING & FILTERING

- Filtering results allows you to focus on particular aspects of your data
- Filter by dragging data fields from the Data pane to the Filters card
- Filtering can also be created by right-clicking a Pill on a Shelf



- In either case, a Filter dialog box will reflect the data type chosen and prompt you for additional criteria



SORTING & FILTERING

Numeric Filter Dialog Box

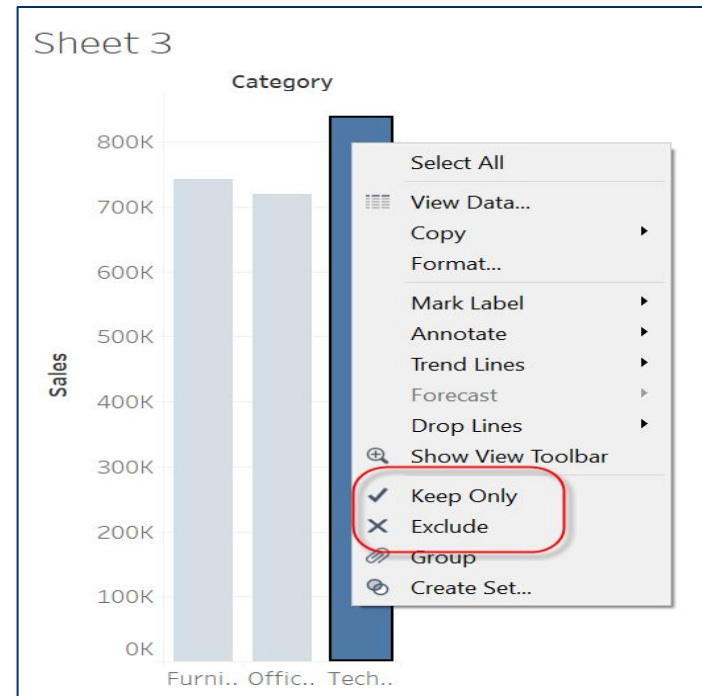
The dialog box is titled "Filter [Sales]". It has four tabs: "Range of values" (selected), "At least", "At most", and "Special". The "Range of values" section shows two input fields with the values "719,047.032" and "836,154.033". Below these fields is a horizontal slider bar with the same values at its ends. At the bottom, there is a "Show:" dropdown menu set to "Only Relevant Values" and an unchecked checkbox for "Include Null Values". The bottom of the dialog features four buttons: "Reset", "OK", "Cancel", and "Apply".

Text Filter Dialog Box

The dialog box is titled "Filter [Customer Name]". It has four tabs: "General" (selected), "Wildcard", "Condition", and "Top". Under the "General" tab, there are three radio buttons: "Select from list" (selected), "Custom value list", and "Use all". Below these is a search box labeled "Enter search text". A list of customer names is displayed with checkboxes next to each: Aaron Bergman, Aaron Hawkins, Aaron Smayling, Adam Bellavance, Adam Hart, Adam Shillingsburg, Adrian Barton, Adrian Hane, Adrian Shami, Aimee Bixby, Alan Barnes, and Max Desrosiers. Below the list are two buttons: "All" and "None", and an unchecked checkbox for "Exclude". A "Summary" section at the bottom shows: Field: [Customer Name], Selection: Selected 0 of 793 values, Wildcard: All, Condition: None, and Limit: None. The bottom of the dialog features four buttons: "Reset", "OK", "Cancel", and "Apply".

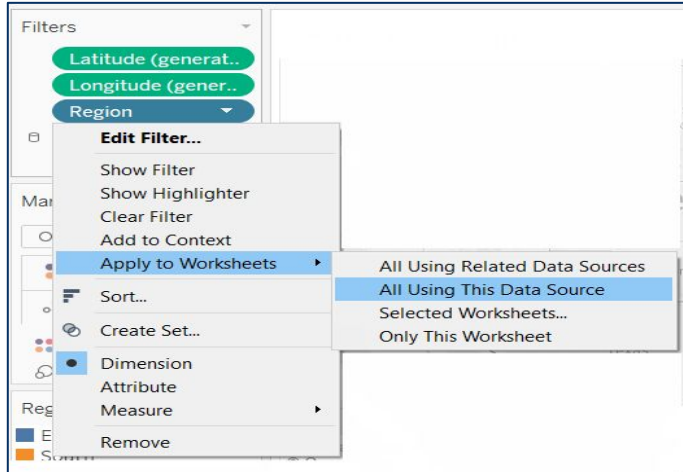
Filtering Data

- Simple filtering can be done by selecting the headers or marks in a View and choosing the **Keep Only** or **Exclude** options
- The Dimension members are removed from the View and the filtered fields are added to the Filters shelf
- When selecting headers that are part of a hierarchy all of the next level members are selected automatically
- Individual marks can be selected on a View, making it easy to focus on or remove outliers



Apply Filters to Multiple Worksheets

- When you had a filter to a worksheet, by default it applies to the current worksheet.
- Sometimes, however, you might want to apply the filter to other worksheets in the workbook.
- You can select specific worksheets to apply the filter to or apply it globally to same data source or related data sources.



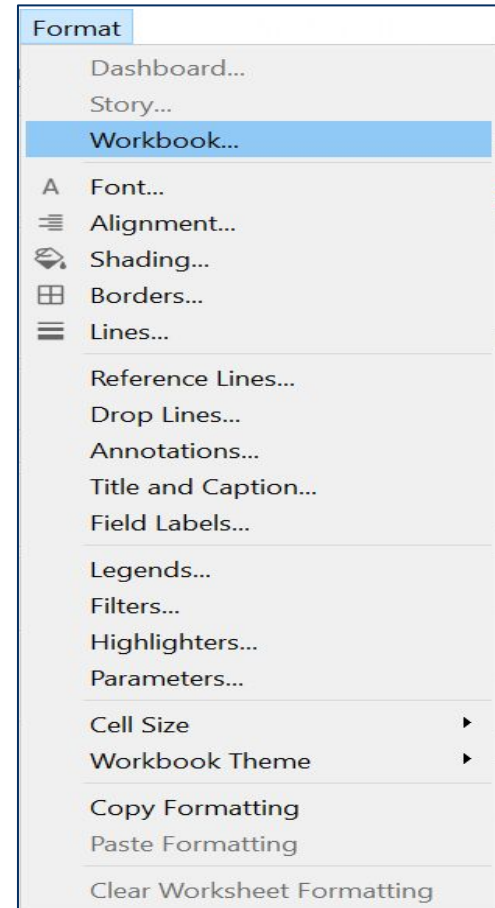
On the Filters shelf, right-click the field and select: **Apply to Worksheets > All Using Related Data Sources**

- Filters that use this option are global across the workbook.



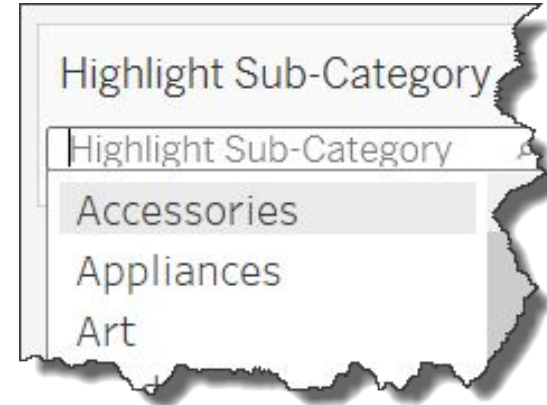
FORMATTING

- Tableau allows for extensive formatting of your views and can be done for an entire worksheet or down to the specific/individual parts.
- There are several ways to initiate formatting in Tableau. You may choose options from the Format drop-down menu or right-click any element and choose **Format** from the context menu.
- Formatting can include changing some or all of the following:
 - Fonts
 - Alignment
 - Shading
 - Borders
 - Lines



Highlighters

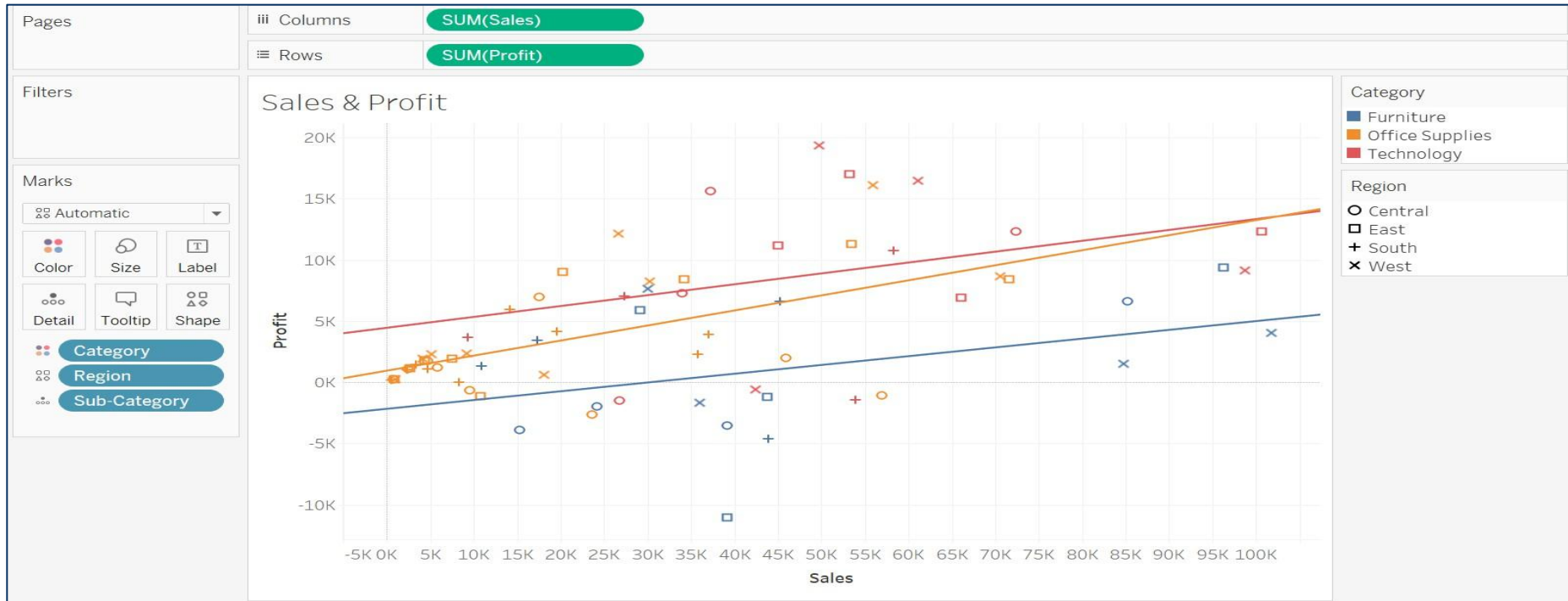
- Highlighters are used to quickly find and highlight data on a visualization
- These come in handy when adding a mark (like color) to a visualization makes it too confusing or hard to read
- Used when there's already a color mark assigned to the visualization
- Previously done through legends and dashboard actions
- Can be used with reference bands and lines
- Highlighter legend supports text searches and dropdown
- Multiple highlighters can be added to a single visualization



The **Marks** card provides other options to customize the appearance of your chart. You can change the color, size, shape, and label of your marks as well as customizing your tooltips.

Tooltips are small pop-up text boxes that appear when you hover your mouse over a mark. By default, tooltips show values for all relevant fields included somewhere on your visualization.





Sum of Sales vs. sum of Profit. Color shows details about Category. Shape shows details about Region. Details are shown for Sub-Category.



BASIC CALCULATIONS

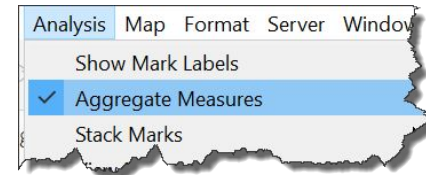
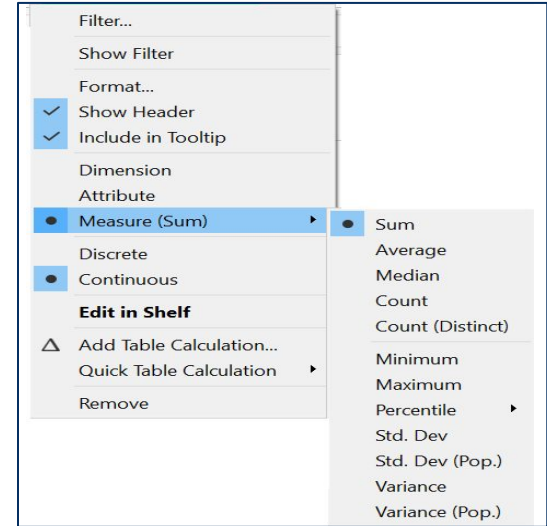
Basic Calculations

- Calculations are created to enhance the dashboard.
- Tableau takes granular data and aggregates them in the View.
- Calculations help in extending the usage of aggregations.
- If we're looking for a particular dimension or measure that does not seem to exist in our data source, we can form them from calculations.

Aggregating Data

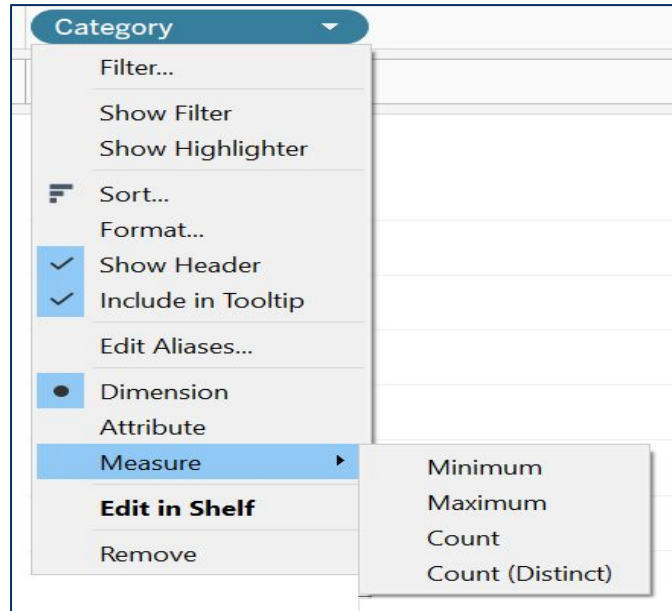
- When a **Measure** is placed on a Shelf, Tableau automatically aggregates the data.
 - Tableau defaults to Sum.
- You can change the aggregation to other types (Avg/Max/Min) by clicking on the Pill, navigating to Measure (Sum) and selecting from the list of other choices

Note: You can disaggregate data by navigating to the **Analysis Menu** and deselecting **Aggregate Measures**.



Aggregating Data

- Dimensions can also be aggregated by right-clicking the Pill, selecting Measure, and choosing from the list of options available:



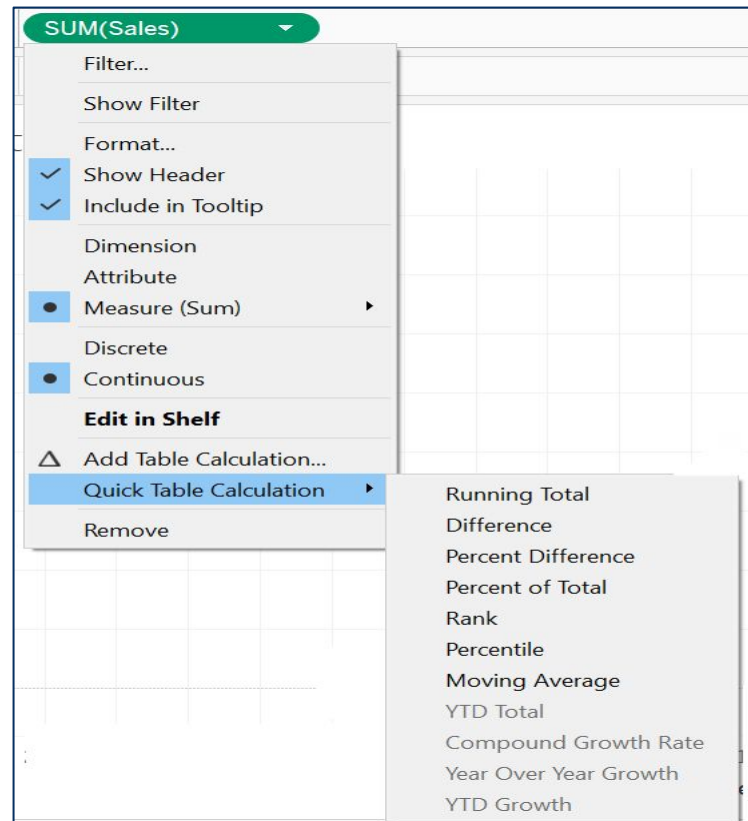
Granularity of Data

- Data granularity refers to the level of detail or depth of data in a View
- Defined by the Dimension fields
- If a Dimension is dragged to Detail, it will change the granularity of the visualization
- Dragging a Measure to Detail will have no effect

Quick Table Calculations

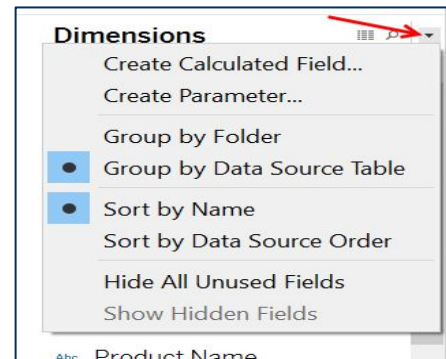
- A quick table calculation is a one-step process where you choose a common, pre-defined calculation type from a list.
- Tableau automatically applies the most typical settings for that calculation type.

To apply a quick table calculation, simply right-click a measure from a Shelf or Mark and choose **Quick Table Calculation** from the context menu.



Calculated Fields

- When your data source doesn't contain all the fields you need, you can add new ones as calculated fields
- The calculated fields are saved as part of the data source and can be used as often as necessary
- There's several ways to create calculated fields. You can do this easily by using the calculation editor from the drop-down, by double-clicking a Shelf or a Field on a Shelf, or by right-clicking "white space" on the Data pane under Dimensions or Measures
- Calculated fields can be created from existing calculations
- Calculations are based on Formulas and Fields



Formulas Make Up Calculated Fields

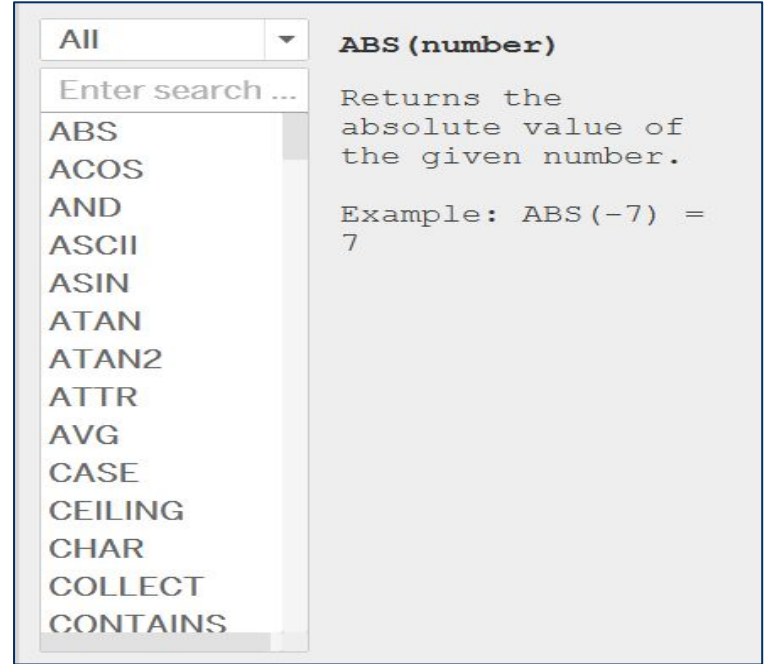
- Formulas can contain functions, fields, operators, parameters, or comments
- The Calculation Editor provides **color coding** based on the Field added
 - Functions: **Light Blue**
 - Fields: **Orange**
 - Operators: **Black**
 - Parameters: **Purple**
 - Comments: **Grey**

✍ **Note:** Tableau makes attempts to guess what you're trying to do and provides you with the tools you need for brackets and parenthesis. You should not have to do much typing or guessing.

*Don't "fight" the tool!

Functions

- Functions can be filtered by category
- Use the **Enter search ...** box to find specific functions
- Clicking on a function provides tips and examples to guide you



The screenshot shows a user interface for selecting functions. At the top, there is a dropdown menu set to 'All'. Below it is a search box labeled 'Enter search ...'. A list of functions is displayed below the search box, including ABS, ACOS, AND, ASCII, ASIN, ATAN, ATAN2, ATTR, AVG, CASE, CEILING, CHAR, COLLECT, and CONTAINS. To the right of the list, the details for the 'ABS (number)' function are shown, including its description and an example.

All ▼

Enter search ...

ABS
ACOS
AND
ASCII
ASIN
ATAN
ATAN2
ATTR
AVG
CASE
CEILING
CHAR
COLLECT
CONTAINS

ABS (number)

Returns the absolute value of the given number.

Example: `ABS(-7) = 7`

Comments

- Comments can provide context or detail on calculations
- Define comments using two forward slashes

Example

```
SUM([Profit])/SUM([Sales])//Added by Amy Such
```

Auto-Completion for Formulas

- Tableau will automatically attempt to fill in a formula
- First attempts to find based on *starts with* then *contains*
- Clicking **Enter** will enter your formula or function into your forming calculation
 - If you first type a [auto-complete will show fields, parameters, or sets instead of functions
- If you're using multiple data sources, auto-complete will add the fully-qualified name of the Field so you can see exactly where things are being pulled from

Objective: We have been asked to show a visualization that displays the average unit cost per each Sub-Category.

1. Create a **New** worksheet. Rename it **Avg Unit Cost**.
2. To find the cost per unit, we need to create a calculated field. Click the drop-down next to the Data pane and select **Create calculated field**.

3. Name the field **Avg Unit Cost**.

4. In the expression box, use the following formula:

[Sales]/[Quantity]

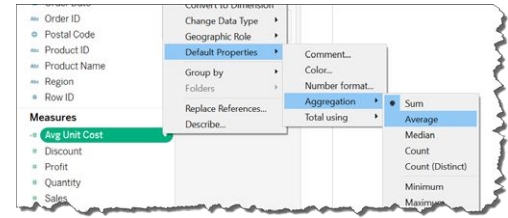


5. Click **OK**.
6. The default aggregation for **Avg Unit Cost** is “Sum”, but this is not what we’re after. We need to see the Average.

7. In the Measures pane, right-click the **Avg Unit Cost** field, and select **Default properties > Aggregation > Average**.

8. Right-click the measure again and select **Default properties > Number Format...** then select **Currency (Standard)**.

9. Click **OK**.

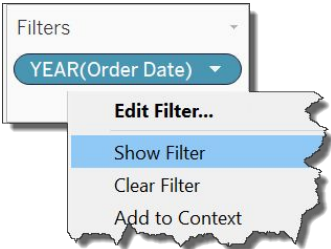


- 10. Add **Sub-Category** to the Columns Shelf and from the Measures pane, drag the newly created **Avg Unit Cost** to Rows.
- 11. Click the **T** button at the top of your screen to add Labels to your visualization.



We would like to be able to select which years to include.

- 12. Drag **Order Date** to the Filters card.
- 13. Select **Years** and click **Next**.
- 14. Select the last 2 years shown in the list.
- 15. Click **OK**.
- 16. Click the down arrow in the Year Filter pill and select **Show Filter** to make it interactive.



OUTCOME



*Average of Avg Unit Cost for each Sub-Category.
The data is filtered on Order Date Year, which keeps 2016 and 2017.*

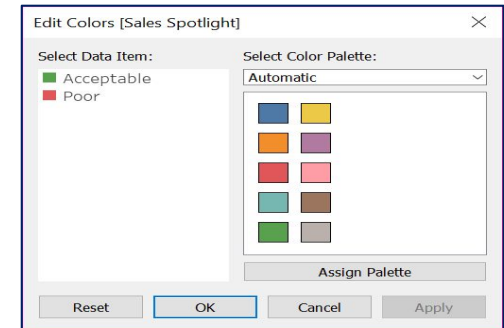
Calculations can be used for exception highlighting. In Tableau, this is known as **Spotlighting**.

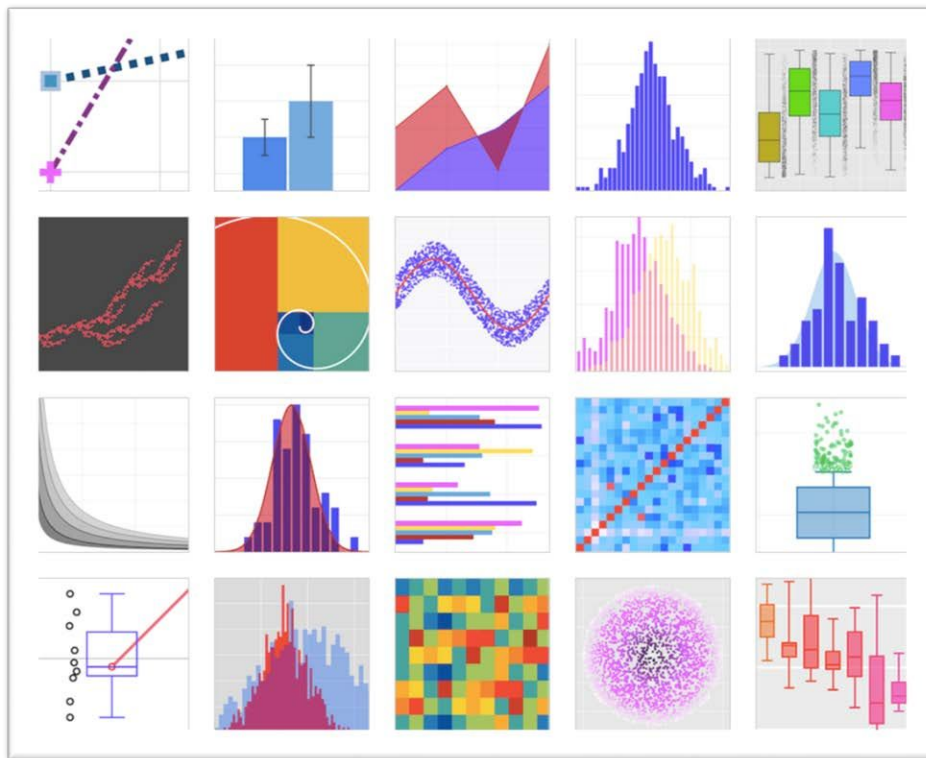
- Spotlighting is based on thresholds set for a selected measure
- It is a technique for showing discrete thresholds based on the values of a measure. For instance, you might want to color-code **Sales** so that those over \$10,000 appear **green** and those below \$10,000 appear **red**.
- A spotlighting calculation is just a special case of a calculation that results in a discrete measure.
 - A discrete measure is a calculation that is a dependent variable (and therefore a measure), but which results in a discrete result (as opposed to a continuous result). Thus the name discrete measure.

SPOTLIGHTING A REPORT

Objective: We have been asked to create a report that highlights when sales are not doing as well as expected. Any order under \$10,000 will be highlighted different from other orders.

1. Create a **New** worksheet. Rename it **Sales Spotlight**.
2. Create a **Calculated field** called **Sales Spotlight**.
3. In the expression box, use the following formula:
IF SUM([Sales])>10,000 THEN "Acceptable" ELSE "Poor" END
4. Click **OK**.
5. Add **Segment** to the Columns shelf and **Category** to the Rows shelf. Also, add **Sub- Category** to the Rows shelf (to the right of Category).
6. For our measure, we will drop **Sales** onto Text in the Marks card.
7. To provide the desired highlighting, drag our newly created **Sales Spotlight** onto Color.
8. Click the drop-down arrow of the Sales Spotlight Legend and select **Edit Colors**.
9. Change "Acceptable" to **Green** and "Poor" to **Red**.





POPULAR VISUALIZATION TYPES

Bar Charts

- Bar Charts are excellent to compare data across categories
- To create a bar chart, add a dimension to the Rows shelf and a measure to the Columns shelf or vice versa
- Adding the dimension (blue) to the Columns shelf and the measure (green) to the Rows shelf will result in a Vertical Bar Chart.

iii Columns	Category
≡ Rows	SUM(Profit)

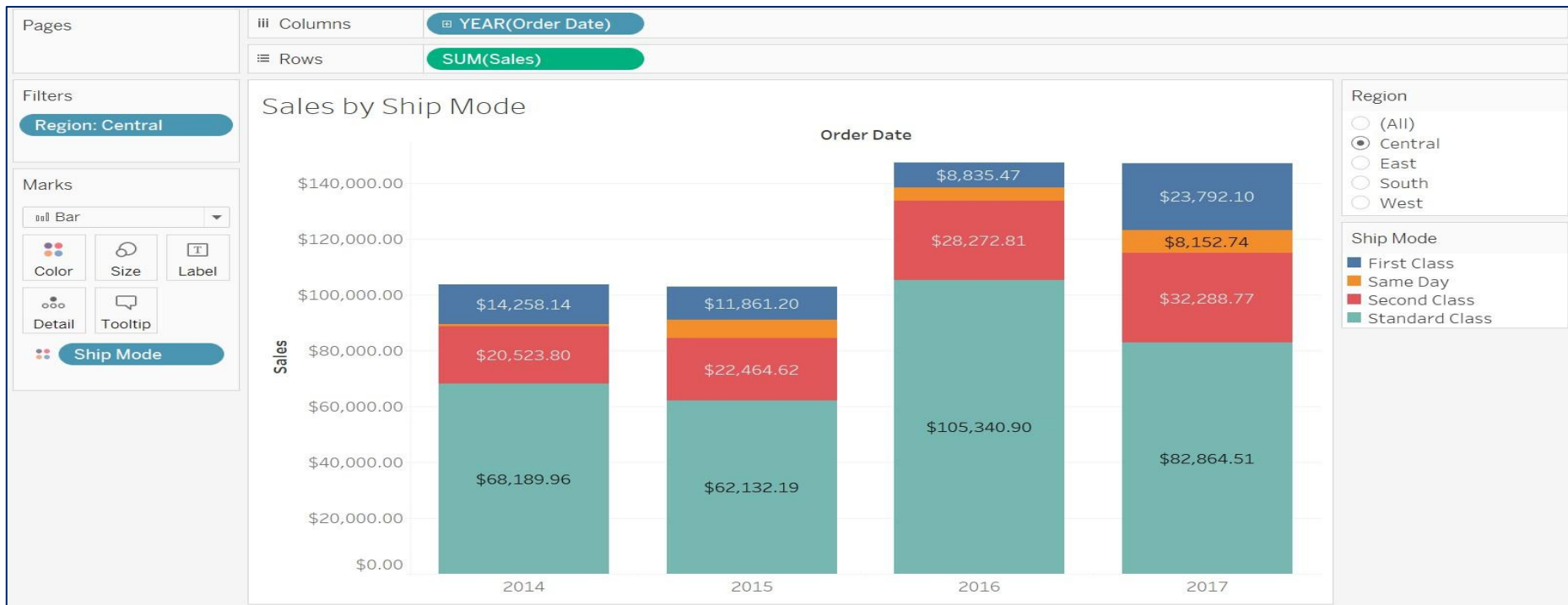
iii Columns	SUM(Profit)
≡ Rows	Category

- Adding the measure (green) to the Columns shelf and the dimension (blue) to the Rows shelf will result in a Horizontal Bar Chart.

STACKED BAR CHART

Objective: We have been asked to create a bar chart that displays total sales over a four year period. In addition, our users want to see the data by shipping methods but only for the West Region.

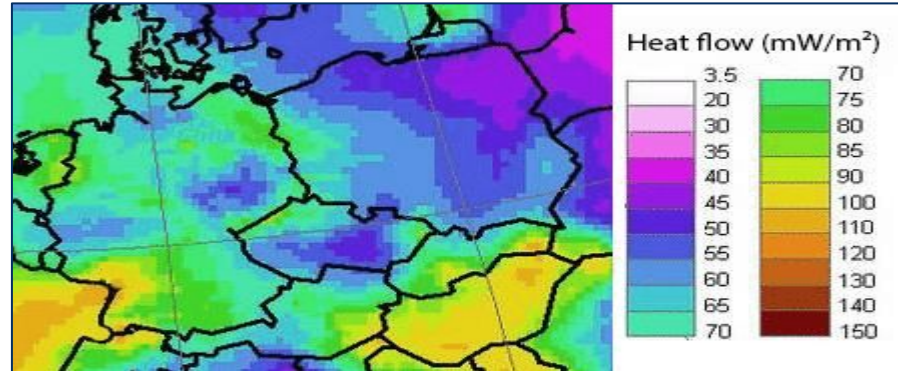
1. Create a **New** worksheet. Rename it **Sales by Ship Mode**.
2. Add **Order Date** to the Columns shelf and **Sales** to the Rows shelf.
Note: Because we have a date and measure on our visualization, Tableau automatically created a line chart. We need to change this.
3. In the Marks pane, change the mark type in the drop-down menu from Automatic to **Bar**.
4. Our next requirement is to show the data by shipping method. To do this drag **Ship Mode** to Color. This changes our bar to a stacked bar.
5. Click the **T** on the toolbar to add Labels to our visualization.
6. Finally, to add to our Filter, right-click **Region** from the Dimensions pane and click **Show Filter**.
7. Click the down-arrow in the Region legend and select **Single Value List**.



*Sum of Sales for each Order Date Year. Color shows details about Ship Mode.
The data is filtered on Region, which keeps Central.*

Heat Maps

- In a Heat map, data is represented in terms of colors. They provide a quick visual summary of the data.
- Heat maps are best used for showing geographical representation of data where each individual value are shown as color.
- Heat maps enable you to compare data in various categories using different colors.

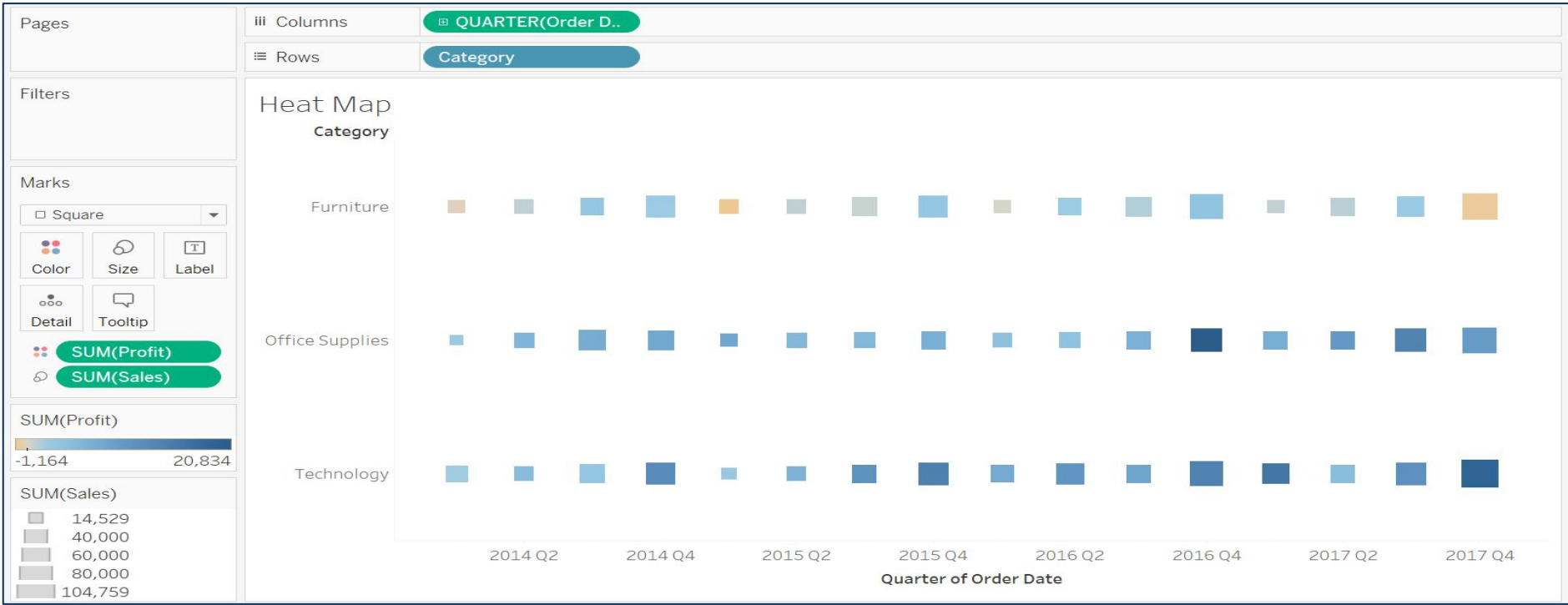


Example: Heat Map

HEAT MAP

Objective: Create a Heat Map to display the Order Date Quarter for each Category.

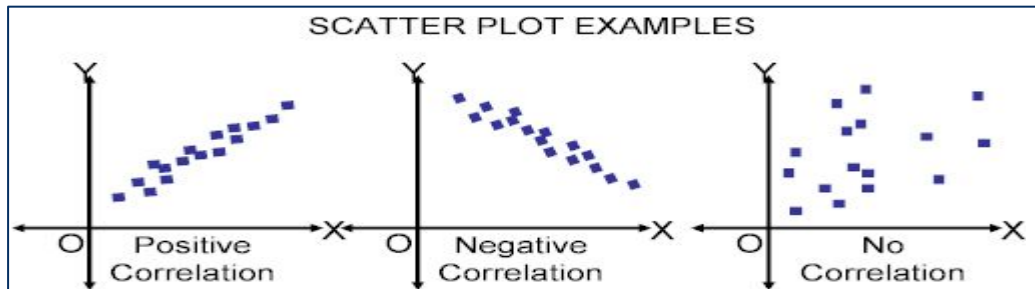
1. Create a **New** worksheet. Rename it **Heat Map**.
2. Drag **Order Date** to the Columns shelf.
3. Expand to show **Quarter**.
4. Right-click the **Year** pill and select **Remove**.
5. Drag **Category** to the Rows shelf.
6. Drag **Profit** to Color.
7. Drag **Sales** to Size.
8. Right-click the **Quarter** pill and select the Quarter from the second tier to show the **Quarter and Year**. This will convert the dimension to a measure.
9. Change the Marks Type to **Square**.
10. Click **Size** and advance the slider to **increase** the size of the marks.



*Order Date Quarter for each Category. Color shows sum of Profit.
Size shows sum of Sales.*

Scatter Plots

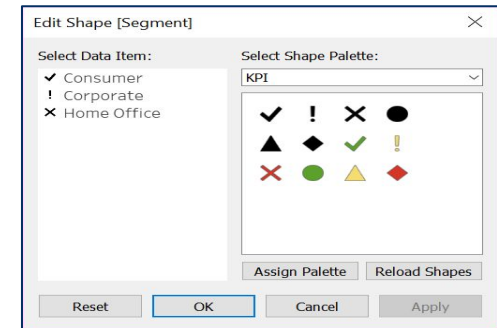
- Scatter Plots can provide a very meaningful visualization of two related numeric measures
- Two different analyses that scatter plots help consider:
 - Comparison/Correlation of the two measures
 - Concentration of data/existence of outliers
- Effective scatter plots include either a small enough number of marks following a general trend to draw quick conclusions about the individual dimensions or a large concentration of marks to draw general conclusions



SCATTER PLOT

Objective: Create a Scatter Plot displaying Sum of Sales vs Sum of Profit.

1. Create a **New** worksheet. Rename it **Scatter Plot**.
2. Drag **Sales** to the Columns shelf.
3. Drag **Profit** to the Rows shelf.
4. Drag **Category** to Color.
5. Drag **Segment** to Shape.
6. Drag **Sales** to Label.
7. Click **Size** and advance the slider to increase the size of the marks.
8. Click the drop-down arrow on the Legend for **Segment** to edit the shapes used in the visualization.
9. Change the shape by selecting **Edit Shape** and assigning the **KPI** palette.
10. Click **OK**.

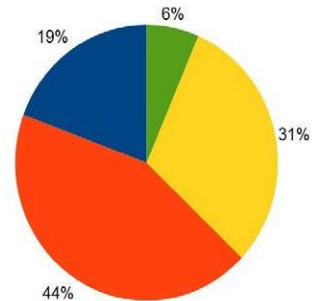




Sum of Sales vs. sum of Profit. Color shows details about Category. Shape shows details about Segment. The marks are labeled by sum of Sales.

Pie Charts

- Do the parts make up a meaningful whole?
 - If not, use a different chart. Only use a pie chart if you can define the entire set in a way that makes sense to the viewer.
- Are the parts mutually exclusive?
 - If there is overlap between the parts, use a different chart.
- Do you want to compare the parts to each other or the parts to the whole?
 - If the main purpose is to compare between the parts, use a different chart. The main purpose of the pie chart is to show part-whole relationships.
- How many parts do you have?
 - If there are more than five to seven, use a different chart. Pie charts with lots of slices (or slices of very different size) are hard to read.



PIE WITH MULTIPLE MARKS

Objective: Create a Create a Pie Chart with Multiple Mark Labels to display Customer Segment, sum of Sales, and % of Total Sales

1. Create a new worksheet. Rename it **Pie with Multiple Marks**.

2. Change the Marks Type to **Pie**.

Notice the “Contextual” Marks card appears for **Angle**.

4. Drag **Segment** to Color.

5. Drag **Segment** to Label.

6. Drag **Sales** to Label.

7. Hover over the **Sales** Pill and click the drop-down arrow.

8. Select **Quick Table Calculation, Percent of Total**.

9. Drag **Sales** to Label (again).

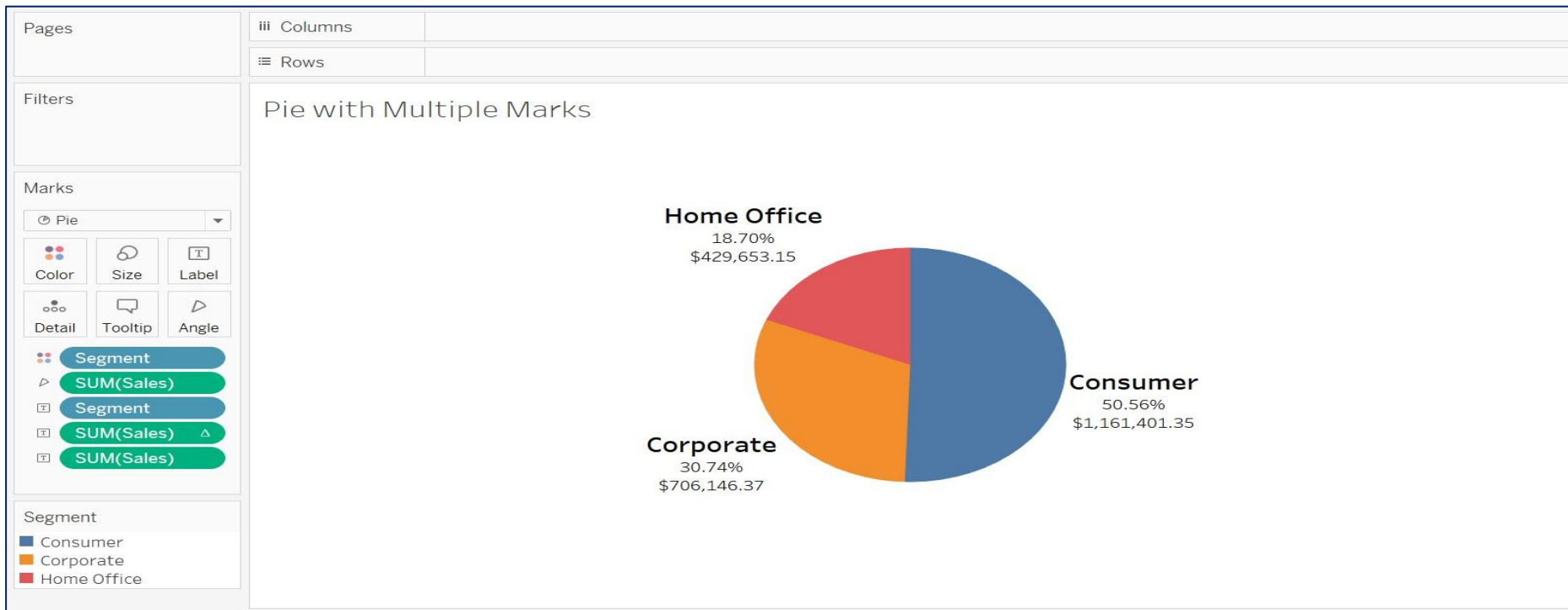
Notice your bar chart still has all equal parts.

10. Drag **Sales** to Angle.

11. Click Label.

12. Click the ellipse (More Button ...) beside Text.

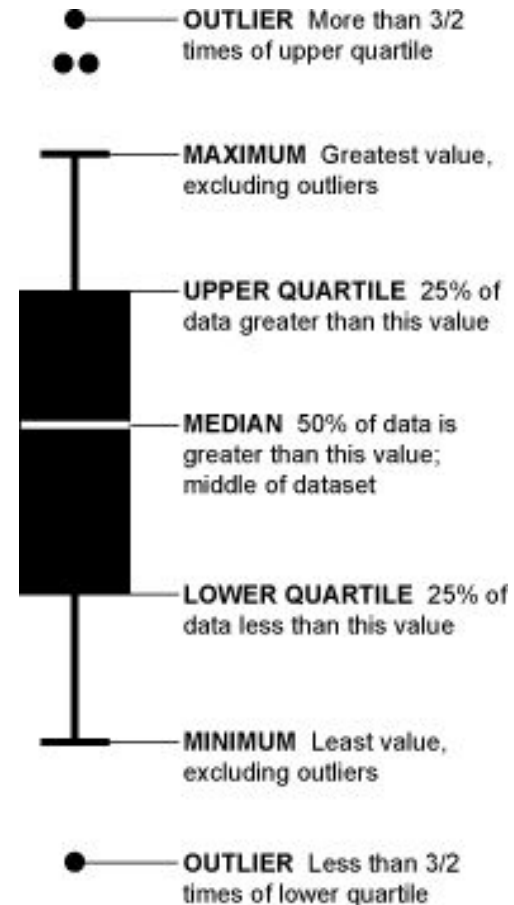
13. Select **<Segment>**, increase the font size and make it bold.



*Segment, % of Total Sales and sum of Sales. Color shows details about Segment.
The marks are labeled by Segment, % of Total Sales and sum of Sales.*

Box Plot

- Also known as Box and Whisker Plot
- Used to show the distribution of data



BOX PLOT



Objective: Create a Box Plot displaying Sum of Profit for each Category.

1. Create a **New** worksheet. Rename it **Box Plot**.
2. Ctrl+Click each of the following: **Category**, **Sub-Category**, **Profit**
3. Expand the **Show Me** Dialog Box.
4. Select the **Box Plot** thumbnail.
5. Drag **Category** to Color.
6. Click the Size Mark and advance the slider to increase the marks of the size on your view.

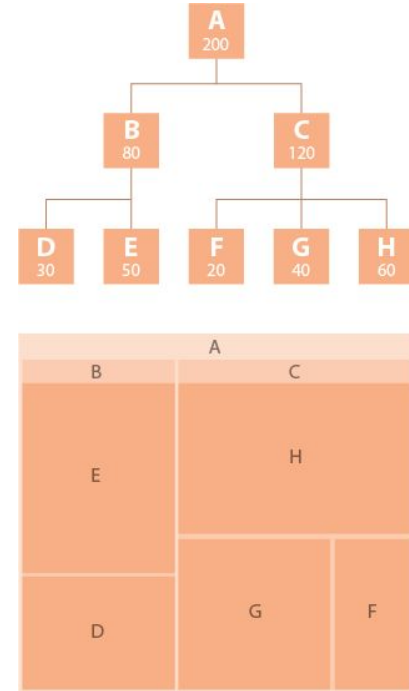




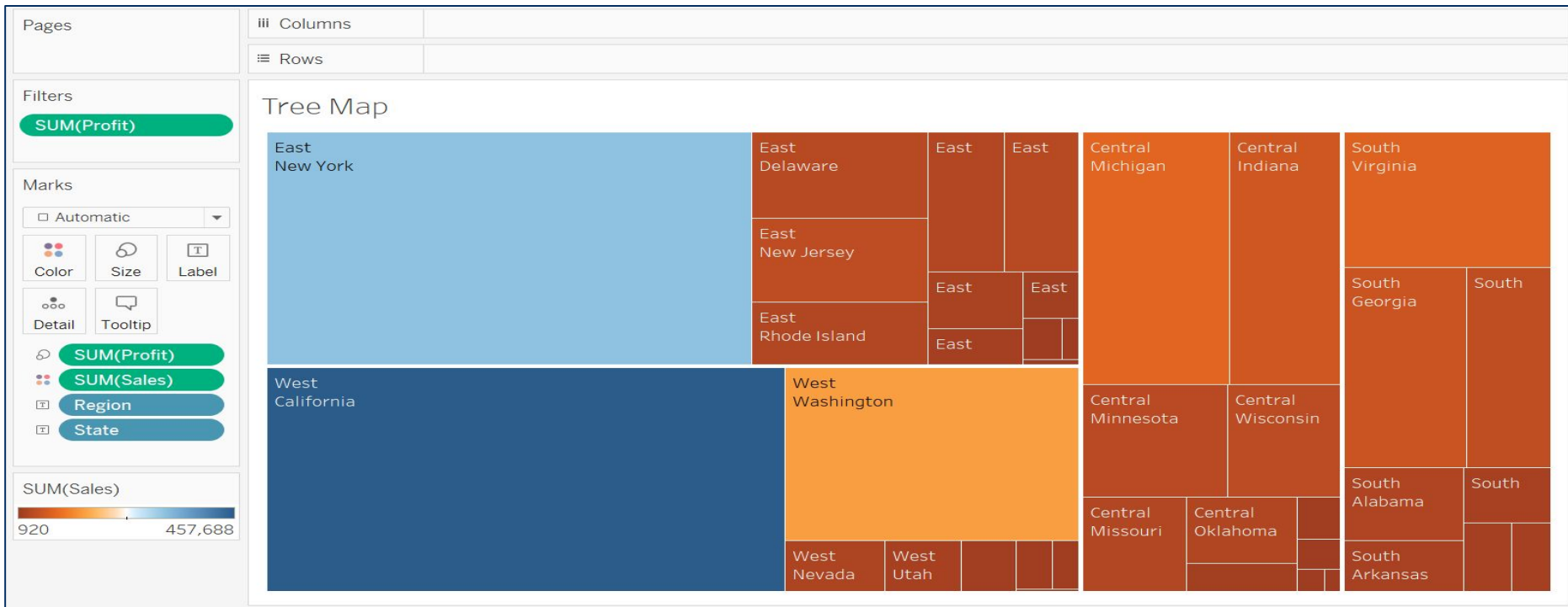
*Sum of Profit for each Category. Color shows details about Category.
Details are shown for Sub-Category.*

Tree Maps

- Treemaps are an alternative way of visualizing the hierarchical structure of a diagram while also displaying quantities for each category via area size
- Each category is assigned a rectangle area with their sub-category rectangles nested inside of it.
- The area size of the parent category is the total of its sub-categories.
- Tree maps start from top left (largest) to bottom right (smallest)



1. Create a **New** worksheet. Rename it **Tree Map**.
2. Ctrl+Click each of the following: **Sales, Profit, Region, State**
3. Expand the **Show Me** Dialog Box.
4. Select the **Tree Map** thumbnail.
5. To edit the colors of your Tree Map, click the drop-down arrow on the legend for Sum of Sales.
6. Select **Edit Colors**.
7. Choose a palette to your liking.
8. Click **OK**.



*Region and State. Color shows sum of Sales. Size shows sum of Profit.
The marks are labeled by Region and State.*

Area Charts

- Area charts are commonly used to showcase data that depicts a time-series relationship
- Unlike Line charts, they can also visually represent volume
- Information is graphed on two axes, using data points connected by line segments. The area between the axis and this line is commonly emphasized with color or shading for legibility
- Typically used when you need to:
 - Know the magnitude of a change
 - Compare a small number of categories
 - Compare between trends and not exact values

